Personalized Medicine Through Personal Data Pods

Elias Crum[1,2], Ruben Taelman[1], Bart Buelens[2], Gökhan Ertaylan[2] and Ruben Verborgh[1]

[1]IDLab, Department of Electronics and Information Systems, Ghent University – imec, Belgium
[2]Flemish institute for Technological Research (VITO) Mol, Belgium

Scalable Personalized Medicine?

€ Unfortunately, still quite expensive

But why? Sequencing is getting pretty cheap.

Answer: Data privacy is crucial
Genomic data storage is expensive

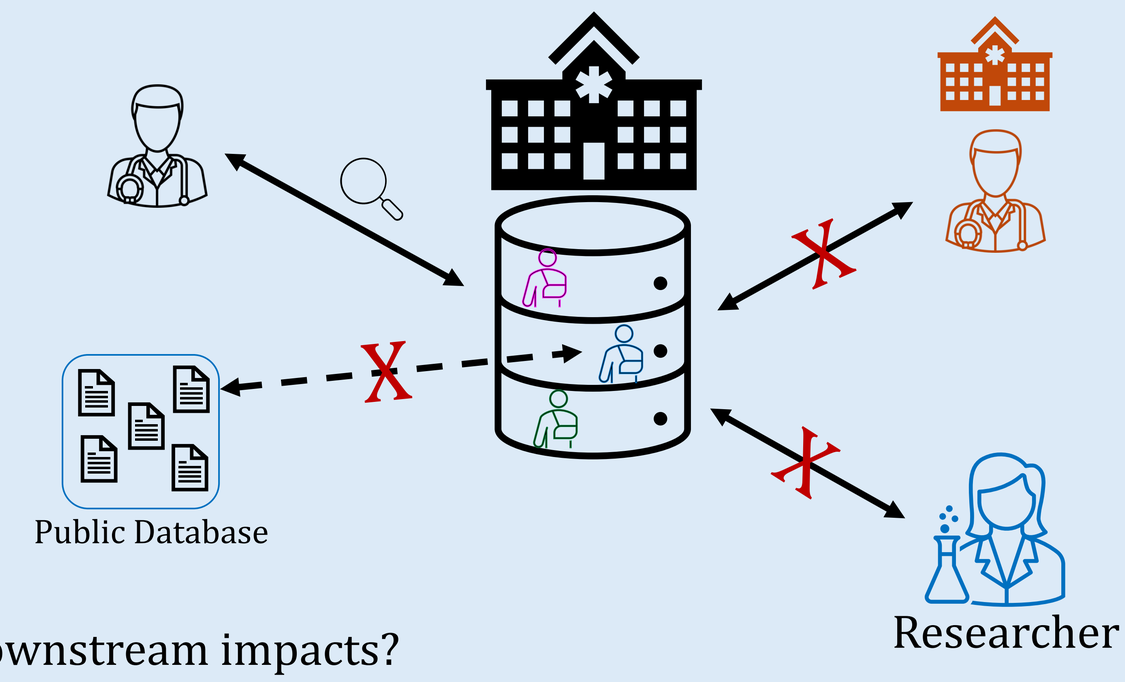
Question: Any way to fix that?

Personal Genome Sequence Data

Very large (~GBs in size)

Require strict privacy protections

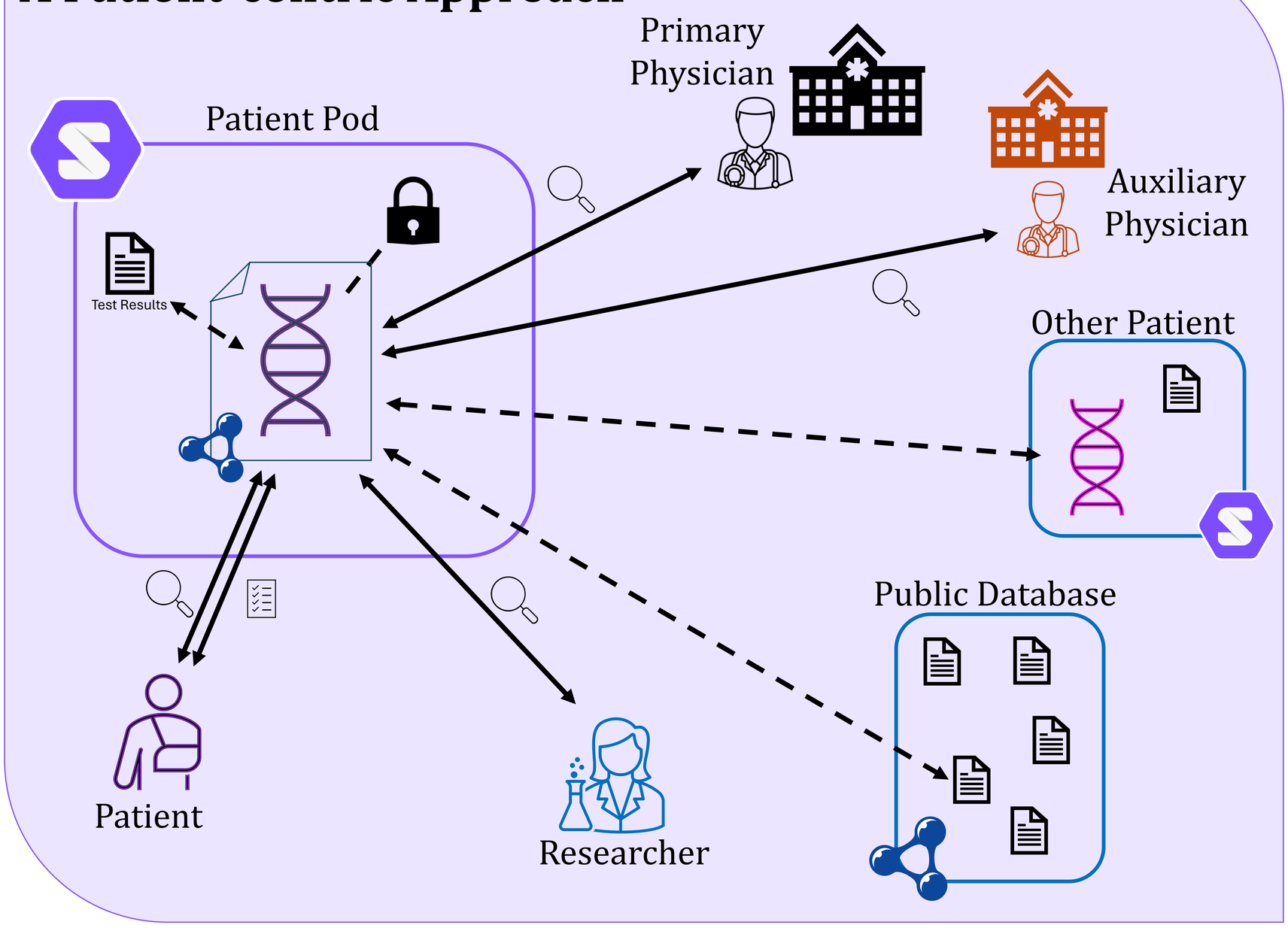
The current institution-centric system



Downstream impacts?

- Reduced data sharing
- Reduced patient transparency
- Increased pertinent data dispersion
- Data duplication

A Patient-centric Approach



Key Challenges

Privacy granularity
- Governance?
- How much patient control?

Data Linkages
- What data should be linked?
- Data formatting Standards?

Querying performance
- Genomic data is BIG
- Requires novel approach?

Poster Paper

Future Directions

Beacon Researcher Access

Consent infrastructure

Unified Framework

Contact Details

Funding provided from VITO NV (UG_PhD_2303_contract)